

Zixin (Jessie) Li

+1 5108165581 | Email | Homepage | LinkedIn

EDUCATION

University of California, Berkeley <i>Master of Arts (MA), Biostatistics — GPA: 4.0/4.0</i>	Aug. 2024 - Present Berkeley, US
Xidian University <i>Bachelor of Science (BS), Mathematics and Applied Mathematics — GPA: 3.9/4.0</i>	Sept. 2020 - Jun. 2024 Xi'an, China
University of California, Berkeley <i>Visiting Student — GPA: 4.0/4.0</i>	Sept. 2022 - Dec. 2022 Berkeley, US

PUBLICATIONS

- Sun, H., **Li, Z.**, Marshall, J. BSAN: Behavioral State Attention Network for Modeling Mosquito Host-Seeking Behavior. *AAAI 2026 AI for Social Impact Track*.
- Li, Z.**, Zhou, R., Cribben, I. Reproducibility in Business Statistics. To be submitted to *The Review of Economics and Statistics*.

AWARDS AND HONORS

Outstanding Graduate Scholarship for Overseas Study <i>Top 1% of students</i>	2024
China Scholarship Council (CSC) Undergraduate Research Award <i>1 of 30 awarded students in China</i>	2023
National Scholarship <i>Top 1% of students</i>	2021
Outstanding Undergraduate <i>Top 1% of students</i>	2021
Meritorious Winner of Interdisciplinary Contest in Modeling (ICM) <i>Top 8% of teams</i>	2021

RESEARCH EXPERIENCE

Large-Scale Human Mobility Generation via Generative AI <i>Research Assistant (Supervisor: Prof. Serina Chang)</i>	Jun. 2025 – Present Berkeley, US
- Build realistic and privacy-preserving synthetic mobility to support population modeling in Richmond, VA, using a large dataset containing 500k+ trajectories and 124 Million+ POI visits. - Developed a novel Generative AI model (BART + DiT) with conditional generation and LLP-style aggregate loss. - Achieved 0.0003 Spatial JSD and 0.005 Travel-Distance JSD, while maintaining strong downstream utility with next-POI prediction accuracy of 0.45 (synthetic) compared to 0.52 (real).	
Multimodal Deep Learning for Mosquito Behavior <i>Research Assistant (Supervisor: Prof. John M. Marshall)</i>	Jun. 2025 – August. 2025 Berkeley, US
- Model mosquito host-seeking behavior from multimodal sensory inputs to support vector-control strategies. - Designed the BSAN framework combining cross-modal attention, recurrent VAE structure, and mixture-of-experts for stochastic flight transitions. - Reduced trajectory error to 0.47 cm ADE and accepted to AAAI 2026 (AI for Social Impact).	
Offline RL for Deep Brain Stimulation <i>Research Assistant (Supervisor: Prof. Lexin Li)</i>	Dec. 2024 – Feb. 2025 Berkeley, US
- Learned DBS control policies from 2 Parkinson's patients with sparse rewards (symptom scores every 2 minutes vs. 120 actions), creating a major credit-assignment gap. - Built a reward-redistribution model to convert trajectory-level BKS/DKS/TS scores into per-step rewards, enabling offline RL with a full simulator and OPE pipeline. - All offline RL algorithms achieved similar performance in simulation; CQL and PARTED were slightly stronger.	

Reproducibility in Statistics and Neuroimaging

Jul. 2023 – Nov. 2023

Research Assistant (Supervisor: Prof. Ivor Cribben)

Edmonton, Canada

- Led a large-scale reproducibility audit of **235 papers** across seven statistics journals to evaluate the reliability of modern computational research.
- Assessed data/code availability and execution pipelines, finding only **17%** reproducible due to missing data (only **37%** provided) and non-functional code (**35%** execution failures).
- Identified systemic reproducibility barriers and improved analytical workflows; additionally applied **local-stationary wavelet models** to fMRI time series to demonstrate rigorous, reproducible dynamic connectivity analysis.

Bandit Learning for Wireless Networks

Dec. 2022 – Apr. 2023

Research Assistant (Supervisor: Prof. Nan Cheng)

Xi'an, China

- Analyze decision-making efficiency of classical bandit algorithms in dynamic wireless systems.
- Benchmarked ϵ -Greedy, UCB, and Thompson Sampling under varying regret and reward conditions..
- Characterized regret convergence behavior and provided guidance for algorithm deployment in dynamic networks.

PROJECT EXPERIENCE

LLM-Guided Reinforcement Learning for Sepsis Management

Apr. 2025

Group Project

Berkeley, US

- Investigated reward sparsity in sepsis treatment using the **MIMIC-III** dataset (**40k ICU stays**) to compare human-defined vs. LLM-generated intermediate rewards for RL policy learning.
- Generated LLM-based clinical assessments from patient trajectories and trained RL agents using Fitted Q-Iteration, evaluated with Step-WIS, MAGIC, and value-based OPE.
- Policies trained with LLM-generated rewards achieved higher estimated value (e.g., **MAGIC = 5.27 vs. 4.91**, **WIS = 4.63 vs. 4.12** for human-defined rewards) and provided more interpretable clinical rationales.

Predicting Pediatric Cervical Spine Injury

Dec. 2024

Group Project

Berkeley, US

- Addressed the challenge of predicting pediatric cervical spine injury using the large PECARN dataset (**3314 patients, 17 hospitals**), where CSI prevalence is **1%** and missed diagnoses carry high clinical risk.
- Built a full pipeline for extensive data cleaning, feature selection, and model development (baseline rule, logistic regression, FIGS tree, neural network).
- The final neural network achieved **0.98 sensitivity**, **0.31 specificity**, and **MMDR = 0.0015**, outperforming the baseline model and reducing unnecessary imaging while maintaining high injury detection accuracy.

TEACHING

PH 142: Introduction to Probability and Statistics in Biology and Public Health

Jul. 2025

Teaching Assistant, UC Berkeley School of Public Health

Berkeley, US

- Assisted in teaching statistical and data science concepts to public health students using R, covering probability distributions and statistical inference.
- Supported lab sessions, graded assignments, and provided guidance on biomedical data analysis and interpretation.

EXTRACURRICULAR ACTIVITIES

Youth Volunteer Association

Jul. 2021

Member

Xi'an, China

- Volunteered as a teaching assistant for children with disabilities, supporting instruction and basic classroom activities.

Interdisciplinary Contest in Modeling (ICM)

Apr. 2021

Meritorious Winner (Top 8%)

- Developed a mathematical model to analyze music influence patterns across genres; ranked in the **top 8%** internationally.

TECHNICAL SKILLS

Python, R, MATLAB, C, L^AT_EX, GitHub